



SmartCart Customer Segmentation System

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Executive Summary

SmartCart is a growing e-commerce platform that struggled with generic marketing strategies and inefficient customer targeting. This project developed an Intelligent Customer Segmentation System using Unsupervised Machine Learning to group customers into meaningful segments and support personalized marketing initiatives

Business Problem

SmartCart lacked visibility into customer behavior patterns, resulting in ineffective campaigns, missed retention opportunities, and difficulty identifying high-value customers.

Project Objectives

- Discover hidden customer behavior patterns
- Segment customers into meaningful groups
- Enable personalized marketing
- Improve customer retention
- Support data-driven decision-making

1. Dataset Overview

```
[2]: df=pd.read_csv('smartcart_data.csv')
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...	MntFishProducts	MntSweetProducts	MntGoldProc
0	5524	1957	Graduation	Single	58138.0	0	0	04-09-2012	58	635	...	172	88	8
1	2174	1954	Graduation	Single	46344.0	1	1	08-03-2014	38	11	...	2	1	
2	4141	1965	Graduation	Together	71613.0	0	0	21-08-2013	26	426	...	111	21	4
3	6182	1984	Graduation	Together	26646.0	1	0	10-02-2014	26	11	...	10	3	
4	5324	1981	PhD	Married	58293.0	1	0	19-01-2014	94	173	...	46	27	1

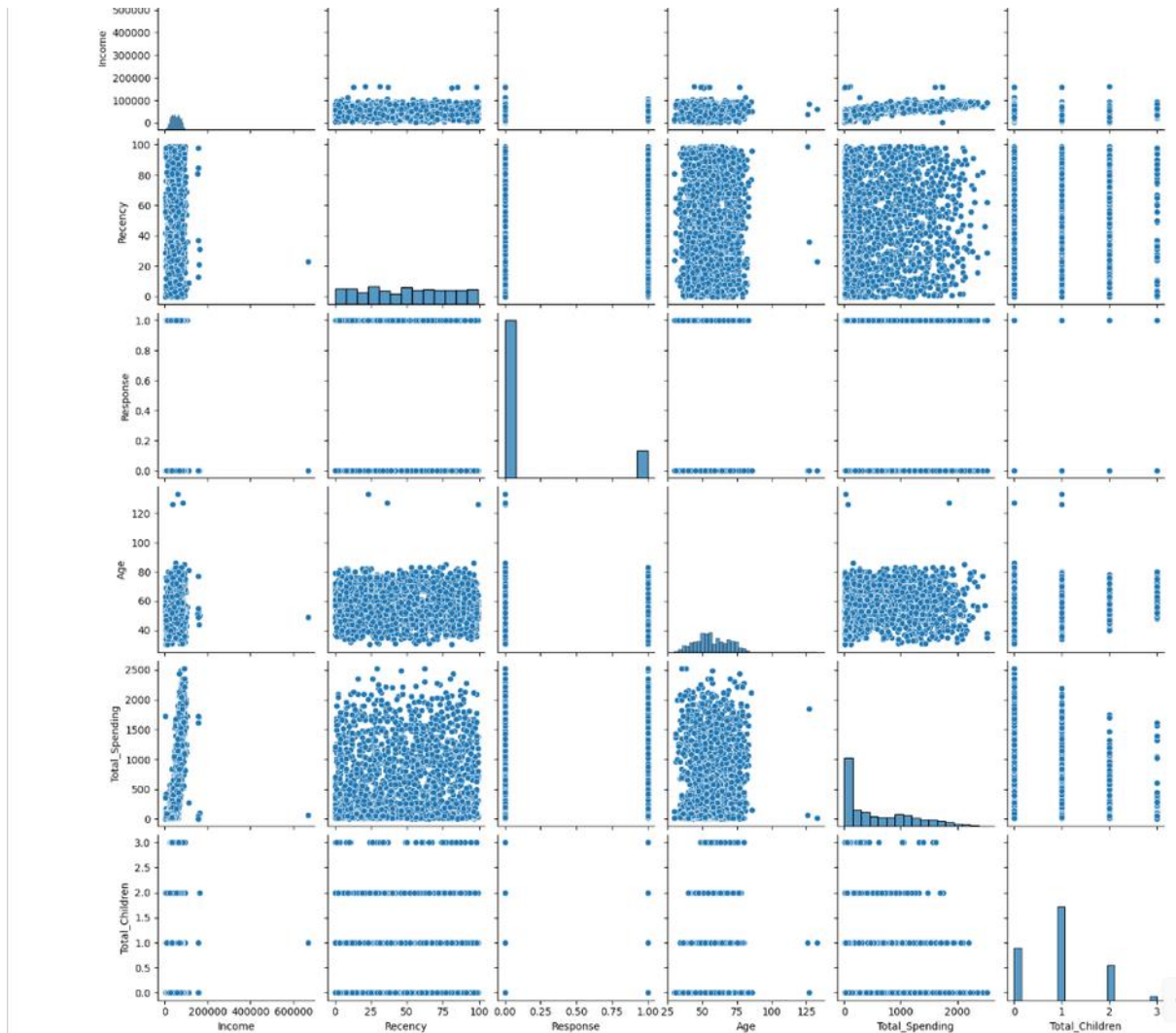
5 rows x 22 columns

```
[3]: df.shape
```

```
[3]: (2240, 22)
```

- The dataset contains 2,240 customer records and 22 features related to demographics, spending behavior, campaign response, and customer engagement.

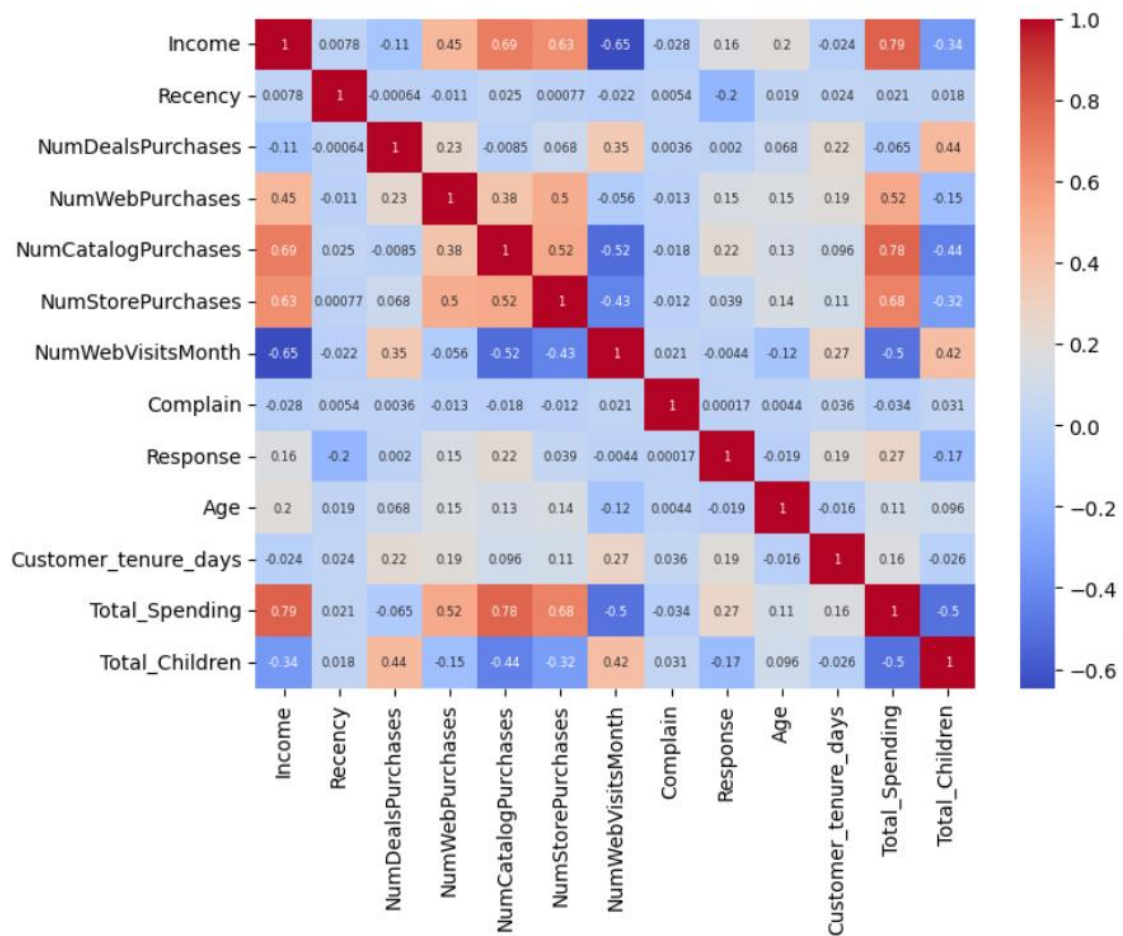
2. Outlier Analysis



- Pair plot analysis revealed the presence of extreme values in Income and Age variables
- After outlier treatment, the dataset size reduced from 2,240 to 2,236 records.

3. Correlation Heatmap

)}: <Axes: >

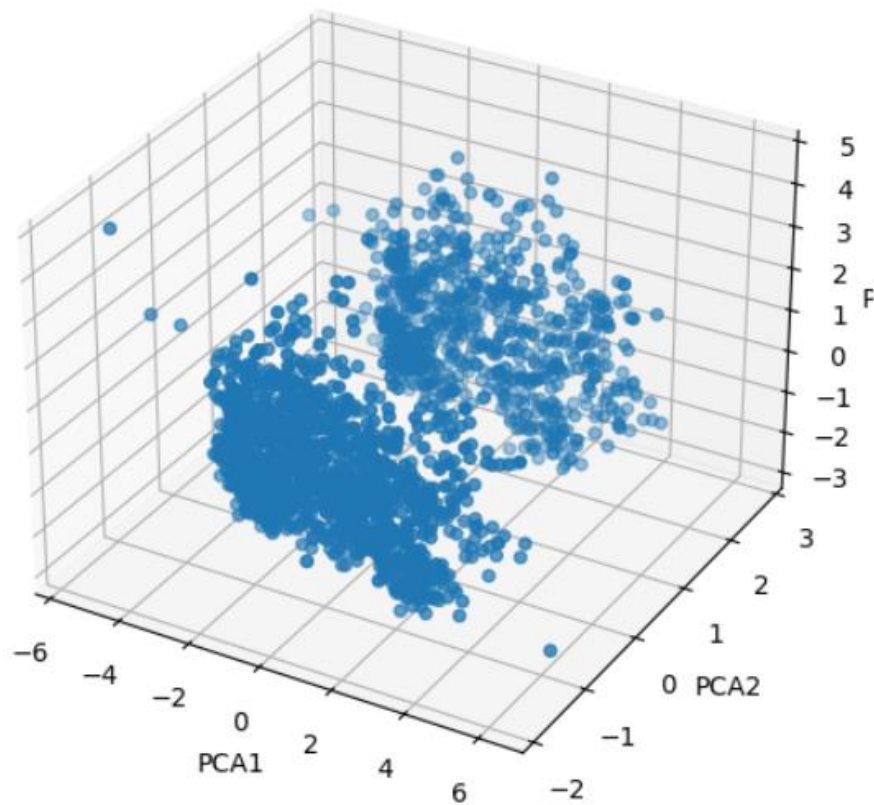


- Income showed a strong positive correlation with Total Spending (0.79).
- Catalog Purchases (0.78) and Store Purchases (0.68) were major contributors to total spending.
- Website Visits (-0.65) showed a negative relationship with Income

4. PCA 3D Projection

: Text(0.5, 0.92, '3D Projection')

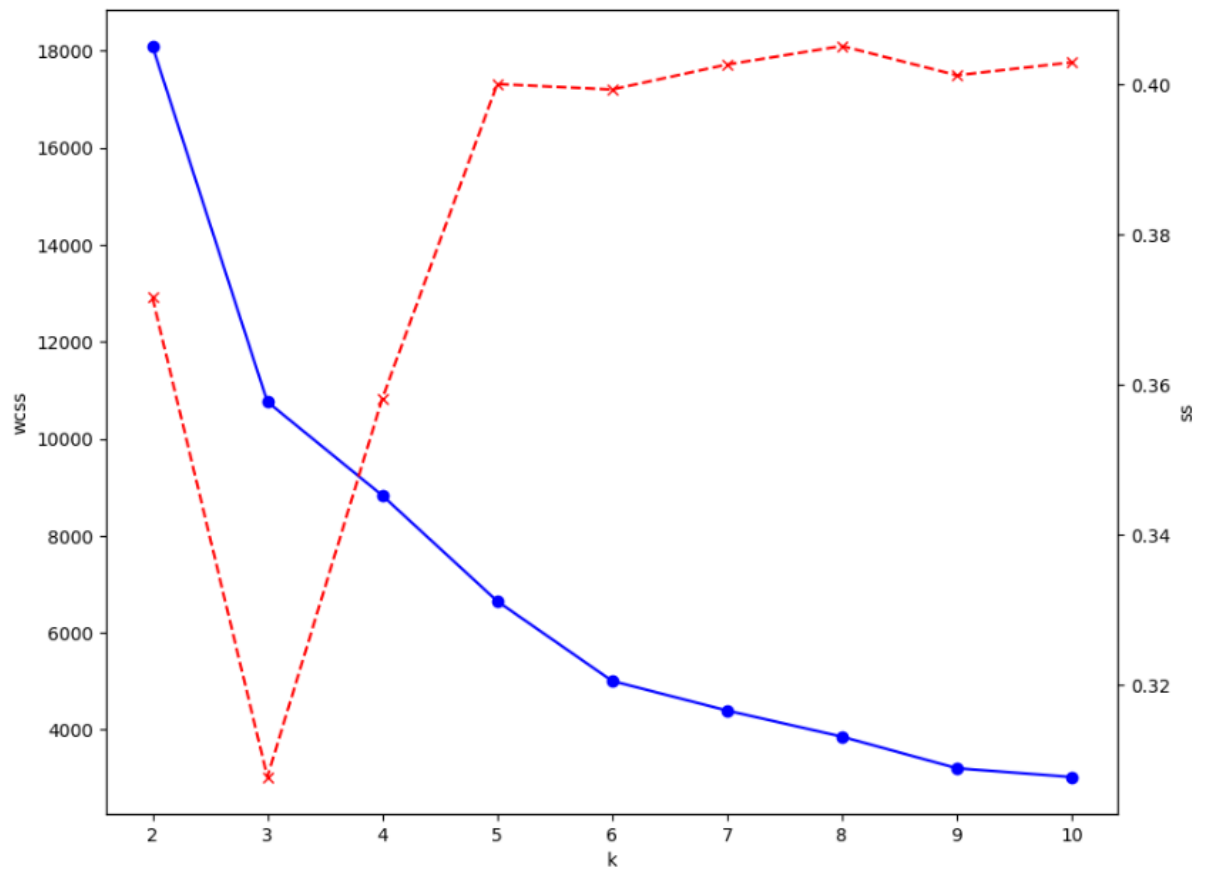
3D Projection



- PCA reduced the dataset into three principal components while preserving customer behaviour patterns.

5. Elbow Method & Silhouette Score Comparison

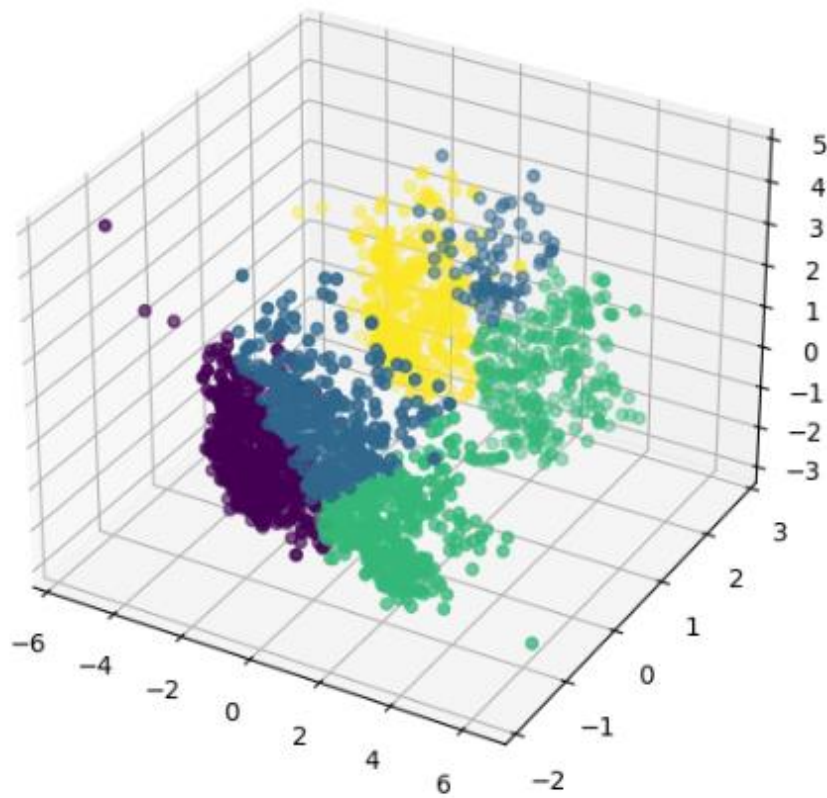
[34]: Text(0, 0.5, 'ss')



- Combining Elbow Method and Silhouette Analysis confirmed 4 clusters as the most suitable segmentation solution.

6. K-Means Clustering

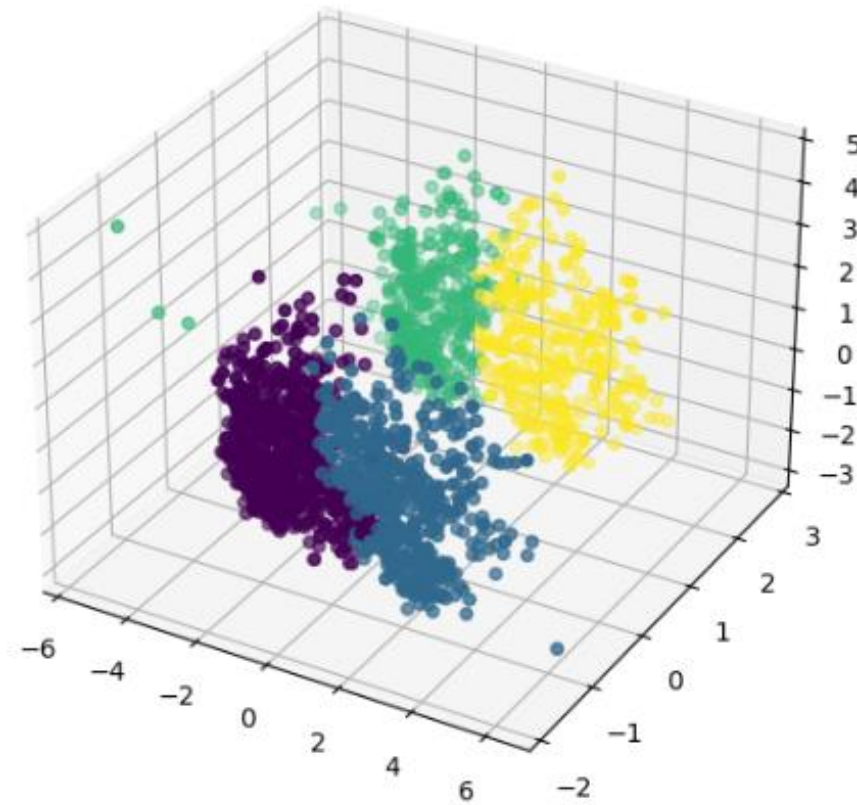
[36]: <mpl_toolkits.mplot3d.art3d.Path3DCollection at 0x1f9cc678670>



- K-Means successfully divided customers into 4 distinct groups based on behavioural and demographic characteristics.

7. Agglomerative Clustering

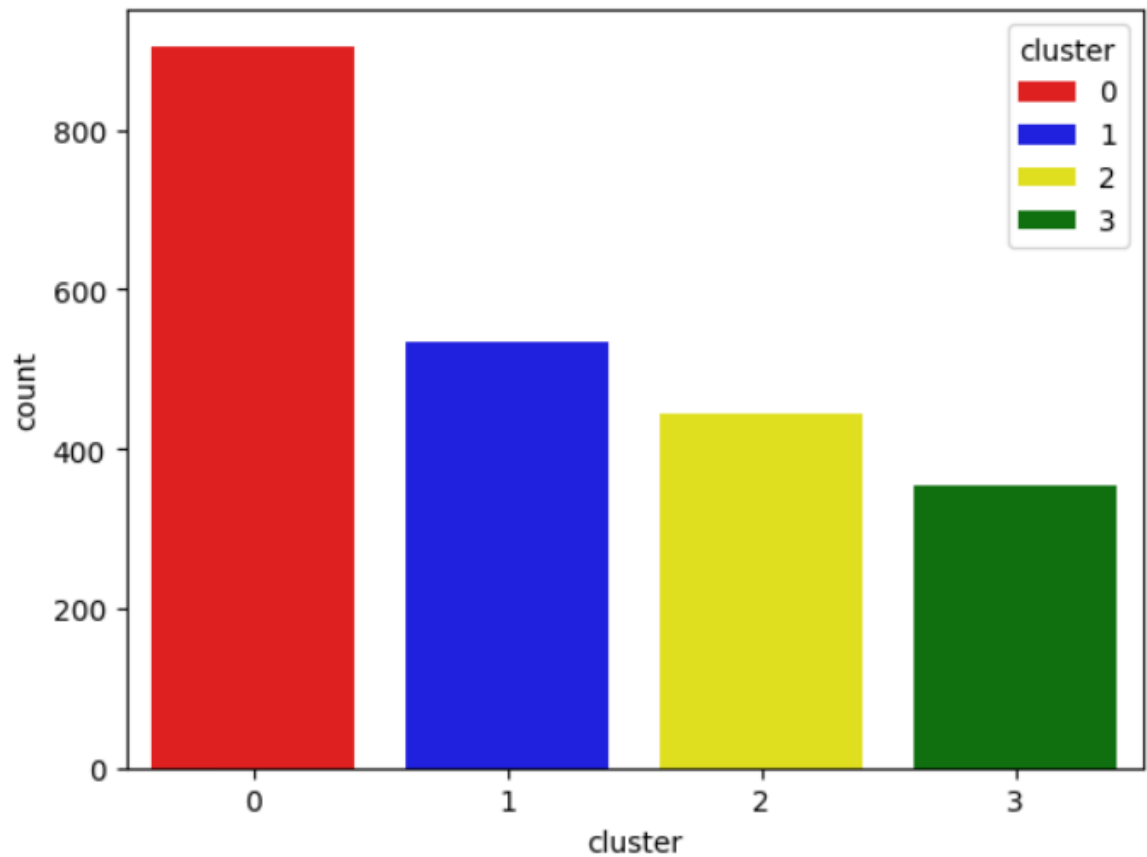
<matplotlib.mplot3d.art3d.Path3DCollection at 0x1f9cc901330>



- Agglomerative Clustering produced 4 well-defined customer segments using hierarchical clustering.
- The model demonstrated stronger cluster separation and customer grouping than K-Means.

8. Customer Distribution Across Clusters

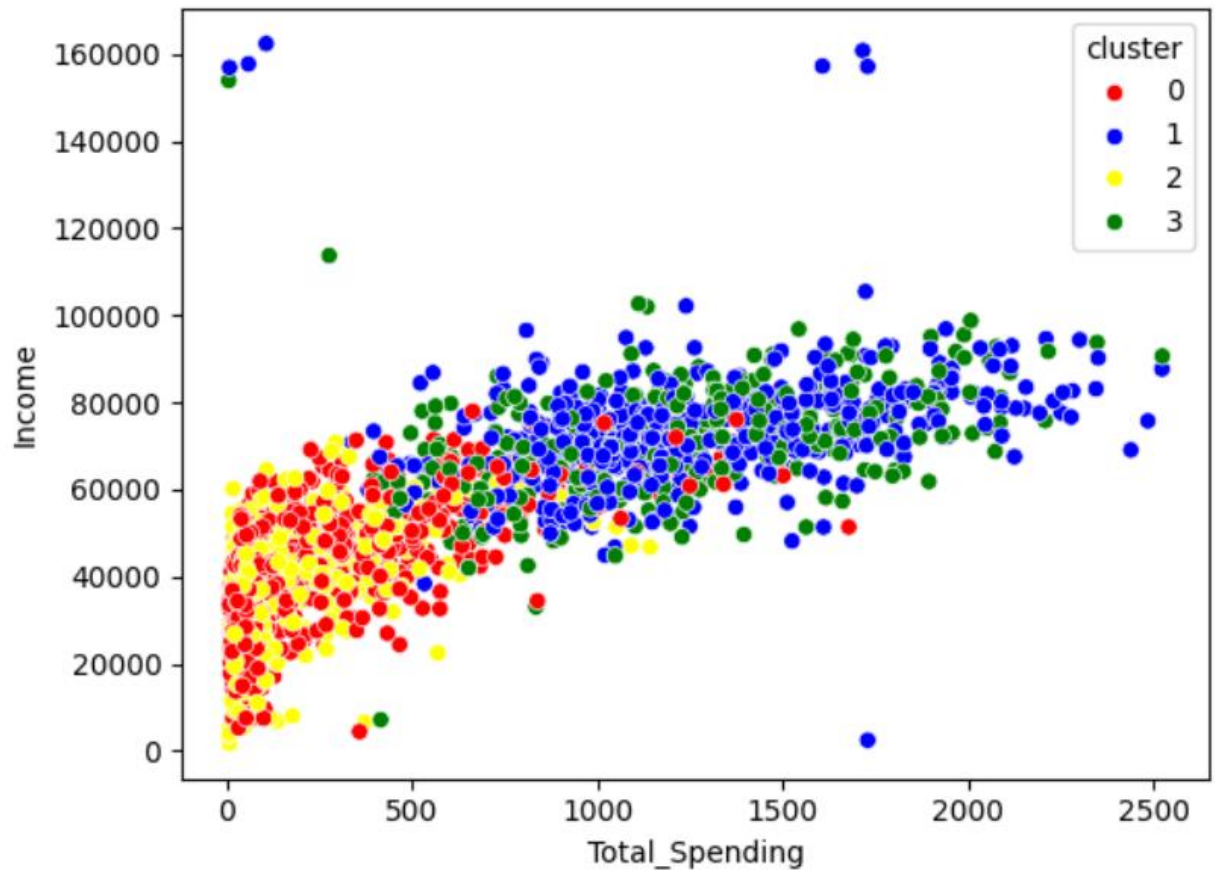
```
[41]: <Axes: xlabel='cluster', ylabel='count'>
```



- Cluster 0 contained approximately 900 customers, making it the largest segment.
- Cluster 1 contained approximately 540 customers.
- Clusters 2 and 3 contained approximately 450 and 350 customers, respectively.

9. Income vs Total Spending Analysis

```
42]: <Axes: xlabel='Total_Spending', ylabel='Income'>
```



- Lower-income customers were primarily grouped into lower-spending clusters.
- Premium customer segments were concentrated in the high-income, high-spending region.